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Open Network for Data Exchange — Technical Documentation

Architecture, identity, transaction flow, and deployment

ATTACHMENT 3 · TECHNICAL REFERENCE

This document describes the architecture of the Open Network for Data Exchange: its design principles, components, identity and signing model, transaction flow, data-protection guarantees, deployment models, and its relationship to Nigeria's national data-exchange standard. It is written for a technical reviewer.

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1. Design principles

- **No central database.** Data remains with the agency that owns it. The network moves questions and signed answers, not records.
- **Verifiable identity.** Every participant holds an asymmetric key pair; every message is signed and every answer verified against a shared registry before it is trusted.
- **Open protocols, no lock-in.** Transport and message structure follow the Beckn protocol; field semantics follow Nigeria's Government Data Exchange (NGDX) standard.
- **Meet agencies where they are.** The same node software runs against a live API, a local database, or a spreadsheet, so the least-digitised agencies can still participate.

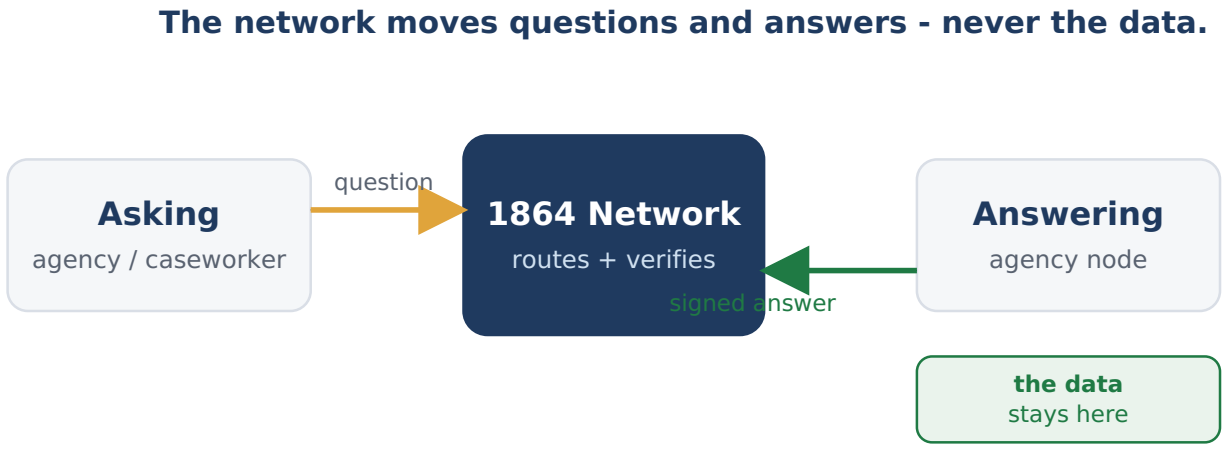


Figure: only a signed answer travels; each agency keeps its records.

2. Components

What runs where

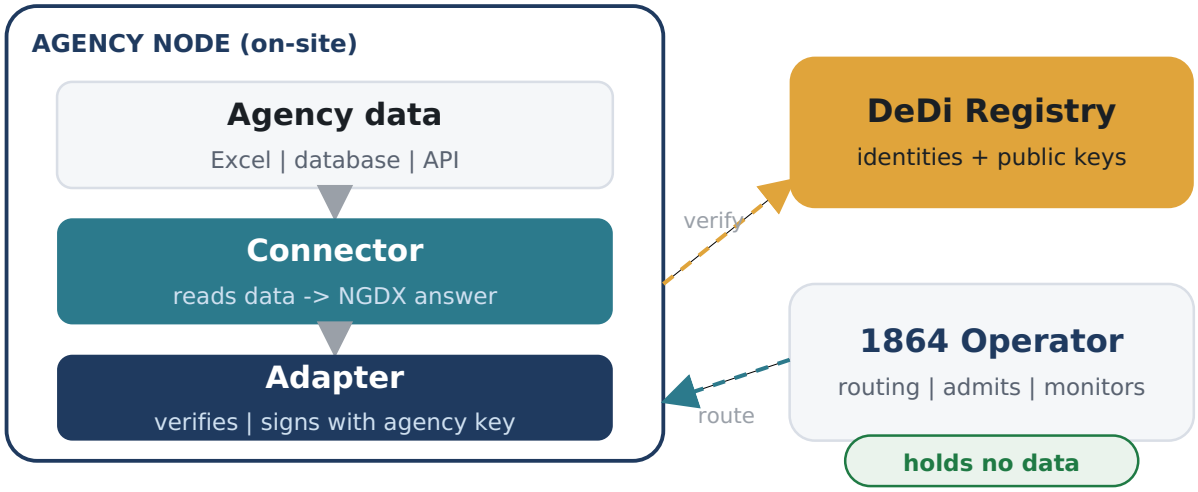


Figure: the node stays with the agency; the operator runs only registry + routing.

Component	Role
Adapter	Protocol middleware at each participant. Verifies inbound signatures, validates message schema, signs outbound messages, and routes them. Both the asking and answering sides run the same adapter; behaviour is set entirely by configuration.
Connector	The thin agency-specific layer behind the adapter. Translates an incoming question into a lookup against the agency's own system (API, database, or spreadsheet) and returns a single scoped answer in NGDX form.
Registry (DeDi)	The decentralised directory of participants. Each entry binds a participant's identifier to its public key, network membership, and callback endpoint. Used for signature verification and routing.
Network operator (1864)	Runs the registry and routing, admits participants, supports onboarding, and monitors uptime. The operator never holds participant data.

3. Identity and signing

Each participant generates an **Ed25519** key pair on its own node; the private key never leaves the machine. The public key, participant identifier, and callback endpoint are registered in the DeDi registry. When a participant sends a message, the adapter signs the message body with the private key. When a participant receives a message, the adapter resolves the sender in the registry, retrieves the sender's public key, and verifies the signature. An answer whose signature does not verify against the registered key is rejected.

Registry lookups resolve a participant by identifier and key reference, returning the signing public key and network membership. Network membership is itself part of the trust boundary: a participant answers only within networks it belongs to, which the operator governs at admission. This is what lets a receiver establish, cryptographically, that an answer came from the claimed agency and was not altered in transit.

Key custody: because private keys are generated and held on each agency's node, the operator cannot sign as an agency. Compromise of the operator does not compromise agency identities.

4. Transaction flow

A verified exchange follows the Beckn request/callback pattern. Using the birth loop as the example:

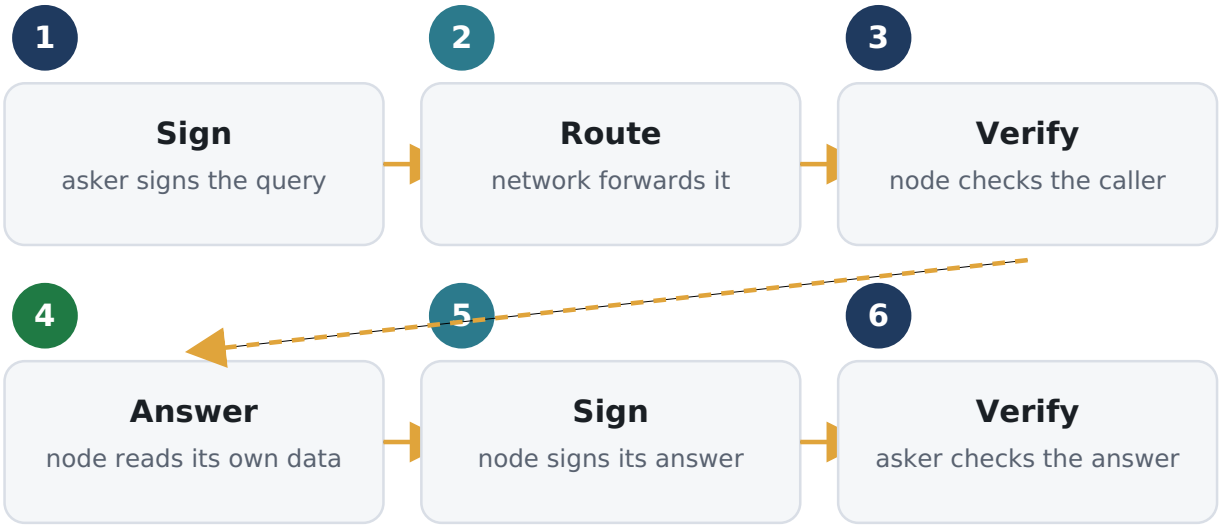


Figure: a verified exchange, from signed question to verified answer.

An illustrative registry verification during a live answer, drawn from the working prototype:

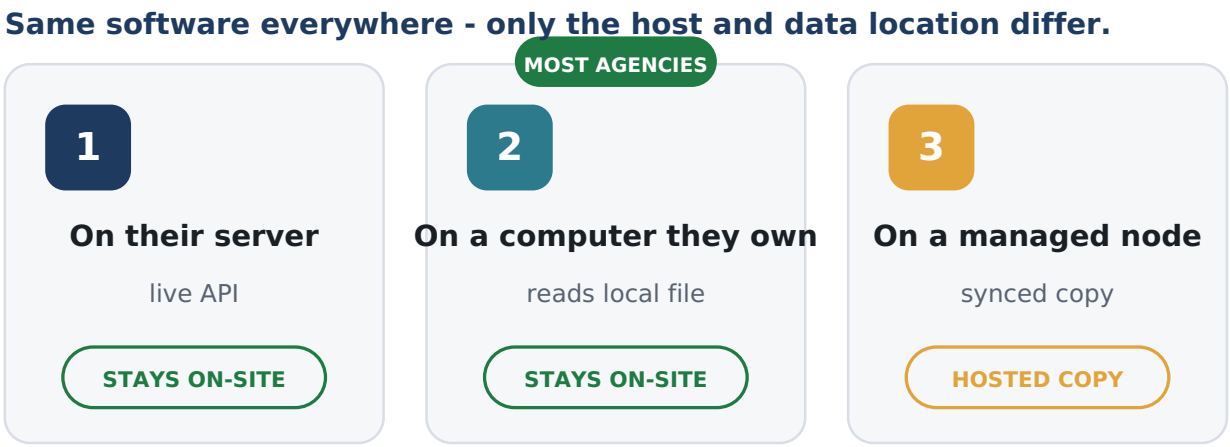
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[RECV]   discover received by npc.node
[LOOKUP] registry resolve caller -> verified
[SIGN]   answer signed as npc.node
[VERIFY] caller verifies signature -> 200 OK
```

5. Data-protection model

- **Scoped answers only.** The connector returns the specific authorised field(s) for a specific question — never a whole record or bulk data.
- **Operator is data-blind.** The network sees message metadata and identities for routing and audit, not the payload data held by agencies.
- **Consent and authorisation.** Participation and permitted queries are governed at the registry and policy layer; the fact of a query is logged for accountability.
- **Local key custody.** Private keys are generated and held on the agency's node; the operator cannot impersonate an agency.
- **Minimisation by construction.** Because only answers move, the volume of personal data crossing the wire is a fraction of any bulk-transfer model.

This model is designed to support compliance with Nigeria's Data Protection Act. Because the operator is verifiably data-blind, an independent review can confirm the property rather than take it on trust; we welcome such a review as part of a pilot.

6. Deployment models



Model	Host	Data source	Data location
Server + API	Agency's existing server	Live API call	On-site, always current
Computer they own	An office machine (or a small provided device)	Local file, or a local DB synced from it	On-site
Managed node	Operator cloud	Synced copy	Hosted — non-sensitive registers only

All models run the same adapter and connector; only the host and data source differ. On-site models require the host to be powered and connected during the agency's chosen active hours. Where mains power is unreliable, a UPS or inverter covering those hours is a normal rollout line item.

7. Relationship to NGDX

NGDX — Nigeria's Government Data Exchange — is the national data rail built on the X-Road model, moving verified records directly between federal agencies with non-repudiable logs. The Open Network is a coordination layer that sits above it: any organisation can discover a service and transact, and can call NGDX to verify a federal fact with consent, but the network runs with or without NGDX present. Adopting the Open Network therefore means implementing a Nigerian standard rather than a proprietary one, and lets sub-federal and non-government participants coordinate immediately, ahead of full national rollout.

8. Current status and roadmap

The standardisation tool (1864 Prep) is live and open source. The Open Network runs end to end in prototype: three Nigerian agency identities (NPC, NIMC, NHIA) are registered with distinct Ed25519 keys, publish to the network, and answer verified queries under their own identities, each verifying successfully. The immediate roadmap:

Stage	What it involves
Pilot design & MoU	One sector, one lead agency, a signed memorandum of understanding, and a defined set of authorised questions.
Agency onboarding	Connectors per agency, node deployment on-site, identity registration, and network admission.
Security & governance	Key-custody procedures, the consent and audit layer, and an independent NDPA review.
Independent evaluation	A randomised evaluation with CEGA at UC Berkeley, changing only the information available, not eligibility or benefits.

Prepared by the 1864 Research and Innovation Institute. For the plain-language overview see the **Open Network User Guide**; for common questions see the **FAQ**.